

PROJECT REPORT

STUDENT NAME : SHAHID ALAM

REGISTRATION NO. : 25BAI11201

UNIVERSITY : VIT BHOPAL UNIVERSITY

Title: Multi-Tenant Campus Hostel & Mess Allocation System

Subtitle: Object-Oriented Java Desktop Application

1. Core Objective & Problem Solved

- Automates campus room assignments and dining plan billing.
- Eliminates manual double-booking errors and billing miscalculations.

2. Key Functional Modules

- Registration & Priority: Calculates student eligibility based on GPA and Academic Year.
- Allocation Engine: Dynamically assigns single/double rooms using preferences and real-time inventory.
- Invoicing & Persistence: Computes total semester fees and saves records to disk.

3. System Architecture (3-Tier)

- UI Layer: Java Swing (MainGUI.java)
- Business Logic: AllocationEngine, BillingService, Student, Room, MessPlan
- Data Layer: File I/O persistence (DatabaseManager.java)

4. How to Compile & Run (CLI)

- Compile:
→ `javac -d bin src/hostel/model/*.java src/hostel/service/*.java src/hostel/ui/*.java src/hostel/Main.java`
- Run App:
→ `java -cp bin hostel.Main`

5. Technical Features & Design Decisions

- Thread Safety: synchronized allocation prevents race conditions during bookings.
- Data Reliability: Auto-saves records to students.txt via Java File I/O.
- Error Handling: Validates numeric fields and alerts users of invalid inputs.

6. Conclusion & Key Takeaway

- **Summary:** Successfully fulfills all academic guidelines by demonstrating OOP principles, modular package design, thread safety, and file persistence.
- **Impact:** Provides a reliable digital solution for managing campus room allocations and billing accurately.